

The Transcription Factor T-bet Regulates Intestinal Inflammation Mediated by Interleukin-7 Receptor+ Innate Lymphoid Cells

SCIFACT-EVIDENCE-20310709

Evidence abstract

Mice lacking the transcription factor T-bet in the innate immune system develop microbiota-dependent colitis.

Here, we show that interleukin-17A (IL-17A)-producing IL-7R (+) innate lymphoid cells (ILCs) were potent promoters of disease in Tbx21(-/-)Rag2(-/-) ulcerative colitis (TRUC) mice.

TNF- produced by CD103(-)CD11b(+) dendritic cells synergized with IL-23 to drive IL-17A production by ILCs, demonstrating a previously unrecognized layer of cellular crosstalk between dendritic cells and ILCs.

We have identified *Helicobacter typhlonius* as a key disease trigger driving excess TNF- production and promoting colitis in TRUC mice.

Crucially, T-bet also suppressed the expression of IL-7R, a key molecule involved in controlling intestinal ILC homeostasis.

The importance of IL-7R signaling in TRUC disease was highlighted by the dramatic reduction in intestinal ILCs and attenuated colitis following IL-7R blockade.

Taken together, these data demonstrate the mechanism by which T-bet regulates the complex interplay between mucosal dendritic cells, ILCs, and the intestinal microbiota.

Document metadata

Corpus | SciFact

Document ID | 20310709